Travel & Tourism Analysis

In [1]:
```
# required library
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as st
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
# load dataset
Travel = pd.read_csv(r'c:\Users\dell\Downloads\Travel And Tourism.csv')
```

In [3]:
```
# read the first 5 rows of the dataset
Travel.head()
```

Out[3]:

	Booking_ID	Booking_Date	Travel_Date	Return_Date	Customer_ID	Customer_Name	Custo
0	1	2026-02-23	2026-04-18	2026-04-21	cust01	Kenzo Vaughn	
1	2	2025-05-30	2025-06-29	2025-07-02	cust02	Priya Carter	
2	3	2024-12-30	2025-03-14	2025-03-23	cust03	Elina Vaughn	
3	4	2026-06-17	2026-09-19	2026-09-26	cust04	Mei Ferraro	
4	5	2025-03-21	2025-04-21	2025-04-24	cust05	Alina Desai	

5 rows × 29 columns

In [4]:
```
# read the last 5 rows of the dataset
Travel.tail()
```

Out[4]:

	Booking_ID	Booking_Date	Travel_Date	Return_Date	Customer_ID	Customer_Name	Cus
995	996	2024-06-10	2024-06-22	2024-06-29	cust996	Sofia Darwish	
996	997	2024-11-28	2025-02-06	2025-02-10	cust997	Ronan Noorani	
997	998	2026-05-18	2026-09-01	2026-09-04	cust998	Lyra Bakshi	
998	999	2024-08-02	2024-10-22	2024-10-27	cust999	Jonah Mori	
999	1000	2025-01-29	2025-05-01	2025-05-06	cust1000	Kai Romero	

5 rows × 29 columns

In [5]:
```
#Information about the dataset
```

```
Travel.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 29 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Booking_ID               1000 non-null   int64  
 1   Booking_Date             1000 non-null   object 
 2   Travel_Date              1000 non-null   object 
 3   Return_Date              1000 non-null   object 
 4   Customer_ID              1000 non-null   object 
 5   Customer_Name            1000 non-null   object 
 6   Customer_Age             1000 non-null   int64  
 7   Gender                   1000 non-null   object 
 8   Customer_City            1000 non-null   object 
 9   Customer_Country         1000 non-null   object 
 10  Destination_City         1000 non-null   object 
 11  Destination_Country      1000 non-null   object 
 12  Number_of_Travellers     1000 non-null   int64  
 13  Number_of_Adults         1000 non-null   int64  
 14  Number_of_Children       1000 non-null   int64  
 15  Hotel_Name               1000 non-null   object 
 16  Hotel_Rating             1000 non-null   float64
 17  Number_of_Rooms          1000 non-null   int64  
 18  Number_of_Nights         1000 non-null   int64  
 19  Meal_Plan                1000 non-null   object 
 20  Payment_Method           1000 non-null   object 
 21  Booking_Status           1000 non-null   object 
 22  Cancellation_Status      1000 non-null   object 
 23  Cancellation_Reason      1000 non-null   object 
 24  Discount_Amount          1000 non-null   int64  
 25  Customer_Rating          1000 non-null   float64
 26  Customer_Review_Comment  1000 non-null   object 
 27  Total_Trip_Cost          1000 non-null   int64  
 28  Transportation_Type      1000 non-null   object 
dtypes: float64(2), int64(9), object(18)
memory usage: 226.7+ KB
```

In [6]:
```
Row = Travel.shape[0]
column = Travel.shape[1]
print(f'The Dataset have {Row} rows and {column} columns\n')
print('List of Columns')

print(Travel.columns)
```

```
The Dataset have 1000 rows and 29 columns

List of Columns
Index(['Booking_ID', 'Booking_Date', 'Travel_Date', 'Return_Date',
       'Customer_ID', 'Customer_Name', 'Customer_Age', 'Gender',
       'Customer_City', 'Customer_Country', 'Destination_City',
       'Destination_Country', 'Number_of_Travellers', 'Number_of_Adults',
       'Number_of_Children', 'Hotel_Name', 'Hotel_Rating', 'Number_of_Roo
ms',
       'Number_of_Nights', 'Meal_Plan', 'Payment_Method', 'Booking_Statu
s',
       'Cancellation_Status', 'Cancellation_Reason', 'Discount_Amount',
       'Customer_Rating', 'Customer_Review_Comment', 'Total_Trip_Cost',
       'Transportation_Type'],
      dtype='object')
```

In [7]:

```
for col in Travel.columns:
    if Travel[col].nunique()<20:
        print(Travel[col].value_counts())
        print('---'*50)
```

```
Gender
Female    568
Male      432
Name: count, dtype: int64
--------------------------------------------------------------------------------
Customer_Country
India                   655
Sri Lanka                50
Canada                   41
United States            39
United Arab Emirates     37
Nepal                    34
Singapore                33
Malaysia                 31
Thailand                 29
Australia                26
United Kingdom           25
Name: count, dtype: int64
--------------------------------------------------------------------------------
Destination_Country
India                   669
Sri Lanka                49
Indonesia                44
United Arab Emirates     42
United Kingdom           41
Thailand                 35
Nepal                    34
Maldives                 32
France                   29
Singapore                25
Name: count, dtype: int64
--------------------------------------------------------------------------------
Number_of_Travellers
5     219
2     156
8     108
4      95
9      93
6      77
1      76
7      43
3      36
12     28
11     26
15     18
13     12
14     12
10      1
Name: count, dtype: int64
--------------------------------------------------------------------------------
Number_of_Adults
5     268
4     201
2     168
3     103
```

```
1      88
6      56
10     41
9      26
8      17
11     15
7      14
12      2
15      1
Name: count, dtype: int64
--------------------------------------------------------------------------------
Number_of_Children
0    481
2    152
3    113
1    111
4    111
5     25
6      6
7      1
Name: count, dtype: int64
--------------------------------------------------------------------------------
Number_of_Rooms
3    297
1    232
4    224
2    136
6     58
5     26
7     26
8      1
Name: count, dtype: int64
--------------------------------------------------------------------------------
Number_of_Nights
8     139
3     137
4     137
5     137
7     130
2     121
6     120
9      43
10     36
Name: count, dtype: int64
--------------------------------------------------------------------------------
Meal_Plan
Breakfast Included    392
Room Only             220
Full Board            208
Half Board            180
Name: count, dtype: int64
--------------------------------------------------------------------------------
Payment_Method
UPI               297
EMI               152
```

```
Credit Card       150
Digital Wallet    149
Debit Card        131
Net Banking       121
Name: count, dtype: int64
--------------------------------------------------------------------------------
Booking_Status
Completed    772
Cancelled    141
Confirmed     79
Pending        8
Name: count, dtype: int64
--------------------------------------------------------------------------------
Cancellation_Status
Not Cancelled    859
Cancelled        141
Name: count, dtype: int64
--------------------------------------------------------------------------------
Cancellation_Reason
Not Applicable               859
Family emergency              27
Transportation disruption     24
Visa issue                    22
Budget constraints            21
Medical emergency             17
Work commitment               16
Change in travel plans        14
Name: count, dtype: int64
--------------------------------------------------------------------------------
Customer_Review_Comment
Trip cancelled before travel             141
Several arrangements need improvement    110
The trip did not meet expectations       102
Poor service experience                  101
Not yet travelled                         87
Enjoyable trip with minor delays          70
Comfortable and well planned              64
Good overall experience                   62
Excellent trip and service                51
Average experience                        46
Everything was well organised             44
Wonderful travel experience               43
Service could be improved                 41
The trip was satisfactory                 38
Name: count, dtype: int64
--------------------------------------------------------------------------------
Transportation_Type
Flight    682
Train     186
Car       132
Name: count, dtype: int64
--------------------------------------------------------------------------------
```

In [8]:
```
# Null values in the dataset
```

```
Null = Travel.isnull().sum().sum()
if Null > 0:
    print('Dataset has null values')
    print('Total NUll is:', Null)
else:
    print('Dataset has no null values')
```

```
Dataset has no null values
```

In [9]:
```
Duplicates = Travel.duplicated().sum()
if Duplicates > 0:
    print('Dataset has duplicate values')
    print('Total Duplicates is:', Duplicates)
else:
    print('Dataset has no duplicate values')
```

```
Dataset has no duplicate values
```

In [10…
```
columns = ['Booking_Date', 'Travel_Date', 'Return_Date']
for col in columns:
    Travel[col] = pd.to_datetime(Travel[col], format = 'mixed', errors =
```

In [11…
```
# Describe of NUmerical columns
Travel.describe().T
```

Out[11]:

	count	mean	min	25%	50%	75%
Booking_ID	1000.0	500.5	1.0	250.75	500.5	750.25
Booking_Date	1000	2025-05-07 07:29:16.800000	2024-01-01 00:00:00	2024-08-31 12:00:00	2025-05-10 12:00:00	2026-01-09 06:00:00
Travel_Date	1000	2025-08-11 03:04:19.200000	2024-01-31 00:00:00	2024-12-11 00:00:00	2025-08-14 00:00:00	2026-04-17 00:00:00
Return_Date	1000	2025-08-16 12:04:19.200000	2024-02-04 00:00:00	2024-12-17 00:00:00	2025-08-18 00:00:00	2026-04-22 00:00:00
Customer_Age	1000.0	34.468	18.0	26.0	34.0	42.0
Number_of_Travellers	1000.0	5.661	1.0	3.0	5.0	8.0
Number_of_Adults	1000.0	4.295	1.0	2.0	4.0	5.0
Number_of_Children	1000.0	1.366	0.0	0.0	1.0	3.0
Hotel_Rating	1000.0	2.9486	1.0	1.8	2.9	4.2
Number_of_Rooms	1000.0	2.959	1.0	2.0	3.0	4.0
Number_of_Nights	1000.0	5.375	2.0	3.0	5.0	7.0
Discount_Amount	1000.0	15980.0	0.0	0.0	7850.0	20125.0
Customer_Rating	1000.0	2.315	0.0	1.0	2.3	3.9
Total_Trip_Cost	1000.0	227838.9	11000.0	81600.0	157200.0	283250.0

Exploratory Data Analysis & Visualizations

In [12…

```
#Summary Statitics of Numerical columns
Num = ['Customer_Age', "Number_of_Travellers", "Number_of_Adults", "Numbe
       'Hotel_Rating', 'Number_of_Rooms', "Number_of_Nights",'Discount_Am
       'Customer_Rating','Total_Trip_Cost']
Summary = Travel[Num].describe().T
Summary['Median'] = Travel[Num].median()
Summary['IQR'] = Summary['75%'] - Summary['25%']
Summary['Skewness'] = Travel[Num].skew()
print('Numberic Summary Statistics:\n')
Summary
```

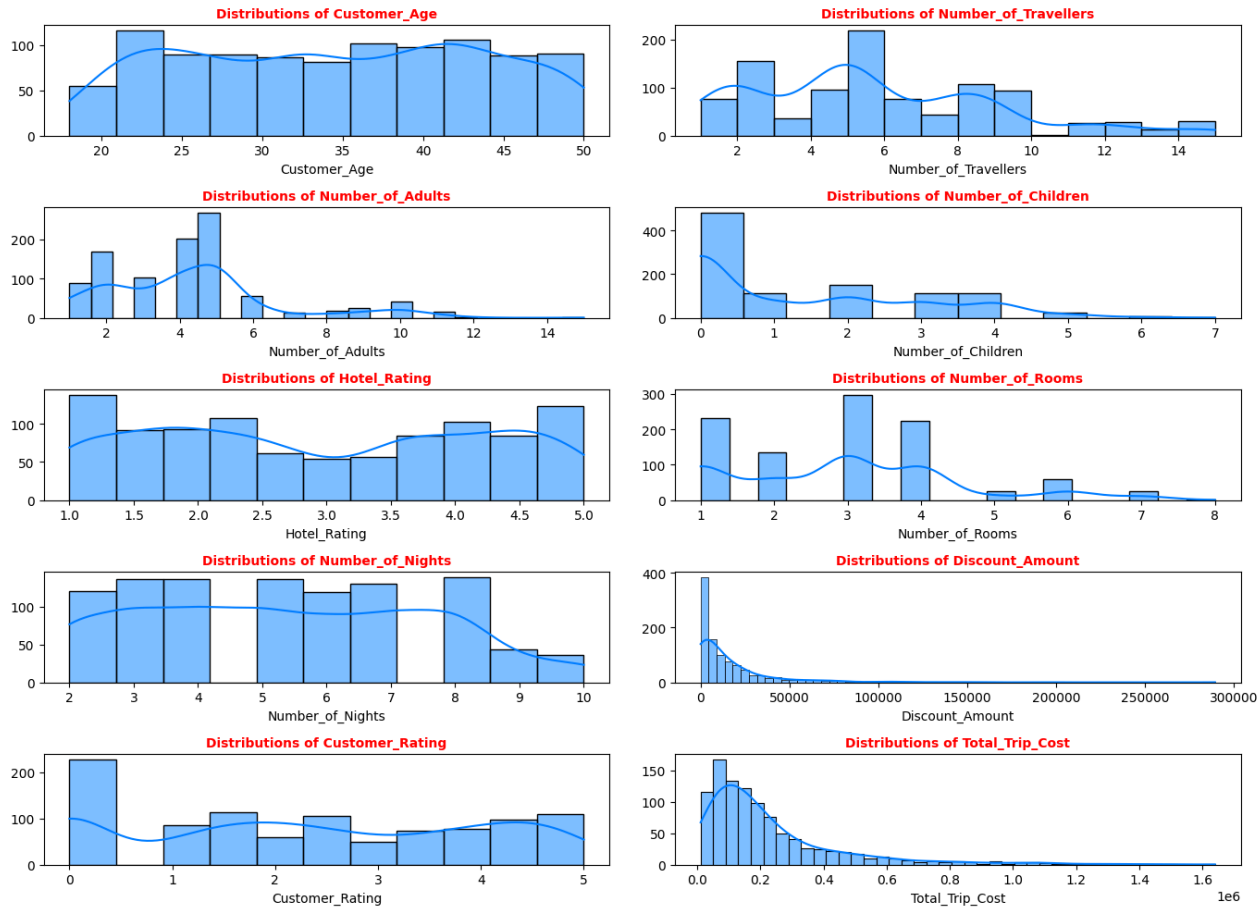
```
Numberic Summary Statistics:
```

Out[12]:

	count	mean	std	min	25%	50%	75
Customer_Age	1000.0	34.4680	9.327914	18.0	26.0	34.0	42
Number_of_Travellers	1000.0	5.6610	3.318900	1.0	3.0	5.0	8
Number_of_Adults	1000.0	4.2950	2.342106	1.0	2.0	4.0	5
Number_of_Children	1000.0	1.3660	1.593293	0.0	0.0	1.0	3
Hotel_Rating	1000.0	2.9486	1.299325	1.0	1.8	2.9	4
Number_of_Rooms	1000.0	2.9590	1.534821	1.0	2.0	3.0	4
Number_of_Nights	1000.0	5.3750	2.262188	2.0	3.0	5.0	7
Discount_Amount	1000.0	15980.0000	24817.542796	0.0	0.0	7850.0	20125
Customer_Rating	1000.0	2.3150	1.673941	0.0	1.0	2.3	3
Total_Trip_Cost	1000.0	227838.9000	226806.646577	11000.0	81600.0	157200.0	283250

In [13…

```
#Numerical Distributions
plt.figure(figsize = (14, 10))
for x, col in enumerate(Num):
    plt.subplot(5,2,x+1)
    sns.histplot(Travel[col], kde = True, color = '#007FFF')
    plt.title(f"Distributions of {col}",
    fontsize= 10, fontweight = 'bold', color= 'red')
    plt.ylabel(' ')
plt.tight_layout()
plt.show()
```



Key insights:

- Age distribution: Most customers are between 25 and 45 years old, with the highest concentration around 30–40. This suggests the market is dominated by working adults and young families.
- Number of travelers: Most bookings involve 2 to 6 travelers, with a strong concentration around 2–4. This supports the idea that trips are mostly for couples or small families.
- Adults vs. children: The number of adults is usually 2–4, while the number of children is mostly 0–2. Very few bookings include large numbers of children, which indicates that family bookings are relatively small and modest.
- Hotel rating: Hotel ratings are concentrated between 3 and 4.5, with a peak around 3.5–4.0. This shows that most travelers book mid-range hotels rather than premium or budget properties.
- Number of rooms: Most stays use 1–4 rooms, which again points to small to medium-sized bookings, consistent with couple/family travel.
- Length of stay: Most trips last 2–6 nights, with a peak around 3–5 nights.

Longer stays are less common, suggesting short to medium tourism trips.

- Discounts: Discount amounts are mostly low, with many bookings having little or no discount, while a few bookings receive very large discounts. This indicates a mostly standard pricing model with occasional promotional offers.
- Customer ratings: Ratings are generally favorable, mostly clustered in the 3.5–4.5 range. This suggests good overall customer satisfaction, with a smaller number of poor experiences.
- Total trip cost: Most trip costs are in the lower-to-mid range, though there are some high-cost outliers. This implies that most packages are affordable to moderately priced, while a minority are high-budget luxury or premium trips.

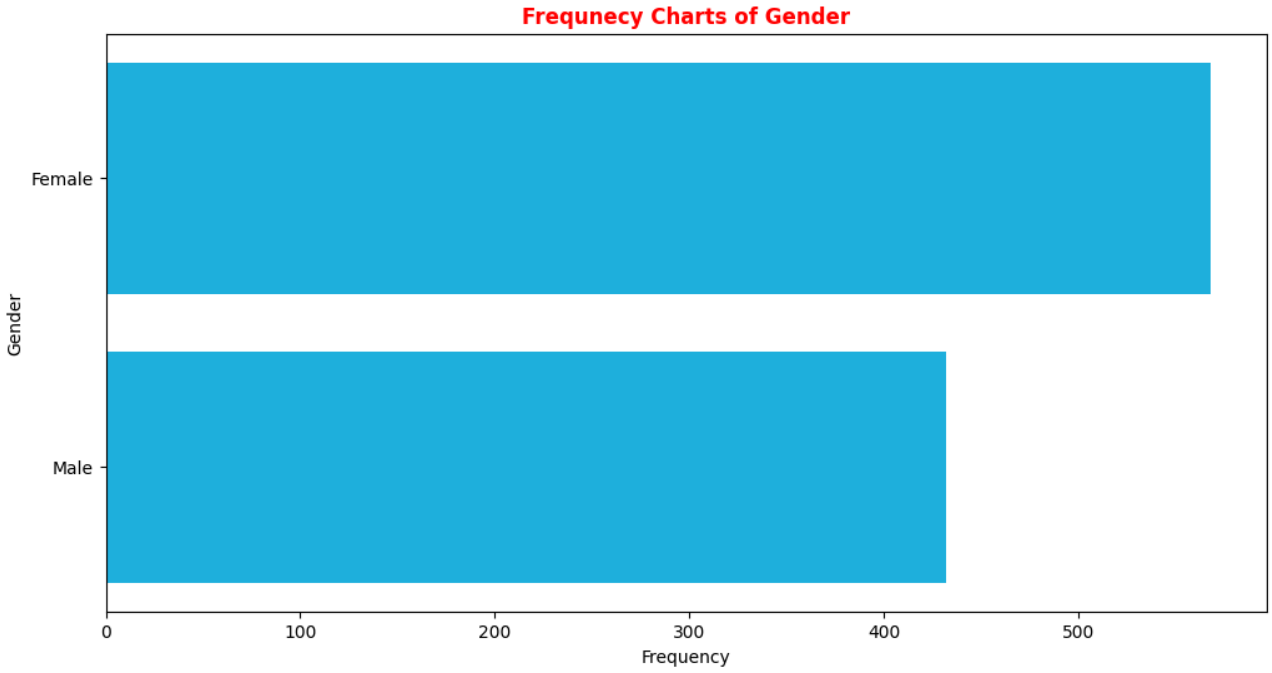
Overall conclusion:

- The dataset is dominated by mid-range, small-group tourism packages.
- The most common customer is a young to middle-aged adult or couple/ family traveler booking a moderate hotel for a short stay at a reasonable price.
- The business appears to serve a broad mainstream tourism segment, not a niche luxury market.

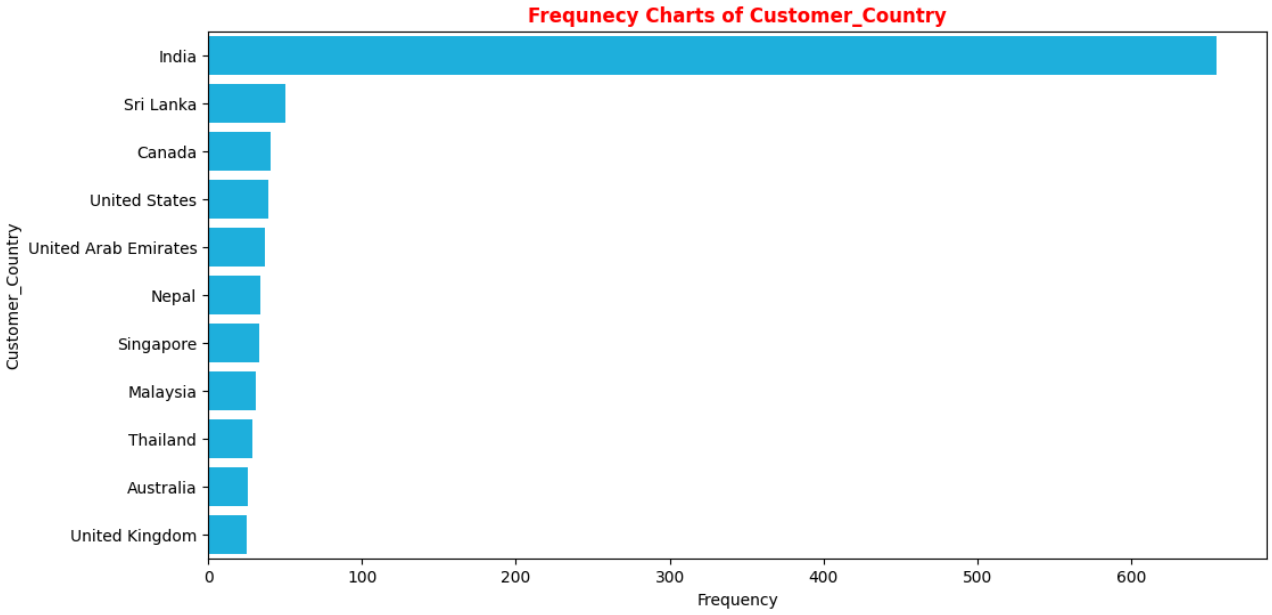
In [14…

```
Category = ['Gender', 'Customer_Country', 'Meal_Plan',
       'Payment_Method', 'Booking_Status', 'Cancellation_Status',
       'Cancellation_Reason','Transportation_Type', 'Customer_City',
       'Customer_Review_Comment']
for x, col in enumerate(Category):
    print(f'\n Frequency of {col}:\n',
    Travel[col].value_counts())
    plt.figure(figsize=(12,6))
    sns.countplot(y= col, data = Travel,
    color = '#00BFFF',order=
    Travel[col].value_counts().index)
    plt.title(f'Frequnecy Charts of {col}',
   fontsize=12,color = 'red', fontweight = 'bold' )
    plt.xlabel('Frequency')
    plt.show()
```

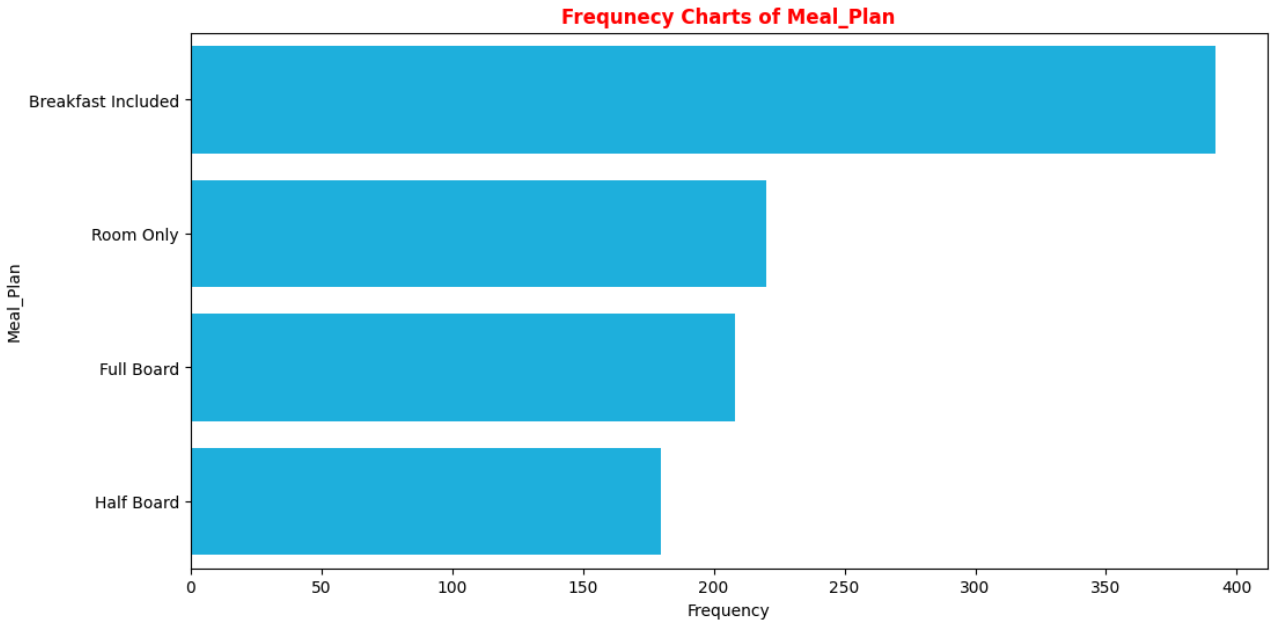
```
 Frequency of Gender:
 Gender
Female    568
Male      432
Name: count, dtype: int64
```



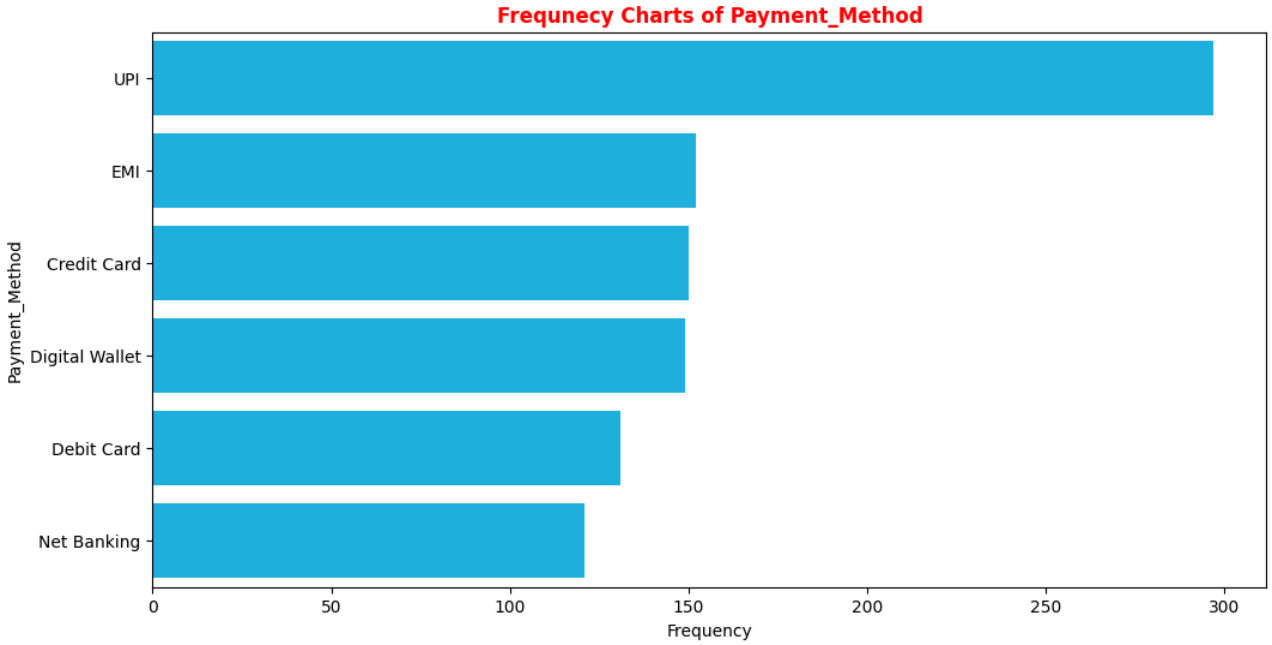

```
 Frequency of Customer_Country:
 Customer_Country
India                    655
Sri Lanka                 50
Canada                    41
United States             39
United Arab Emirates      37
Nepal                     34
Singapore                 33
Malaysia                  31
Thailand                  29
Australia                 26
United Kingdom            25
Name: count, dtype: int64
```



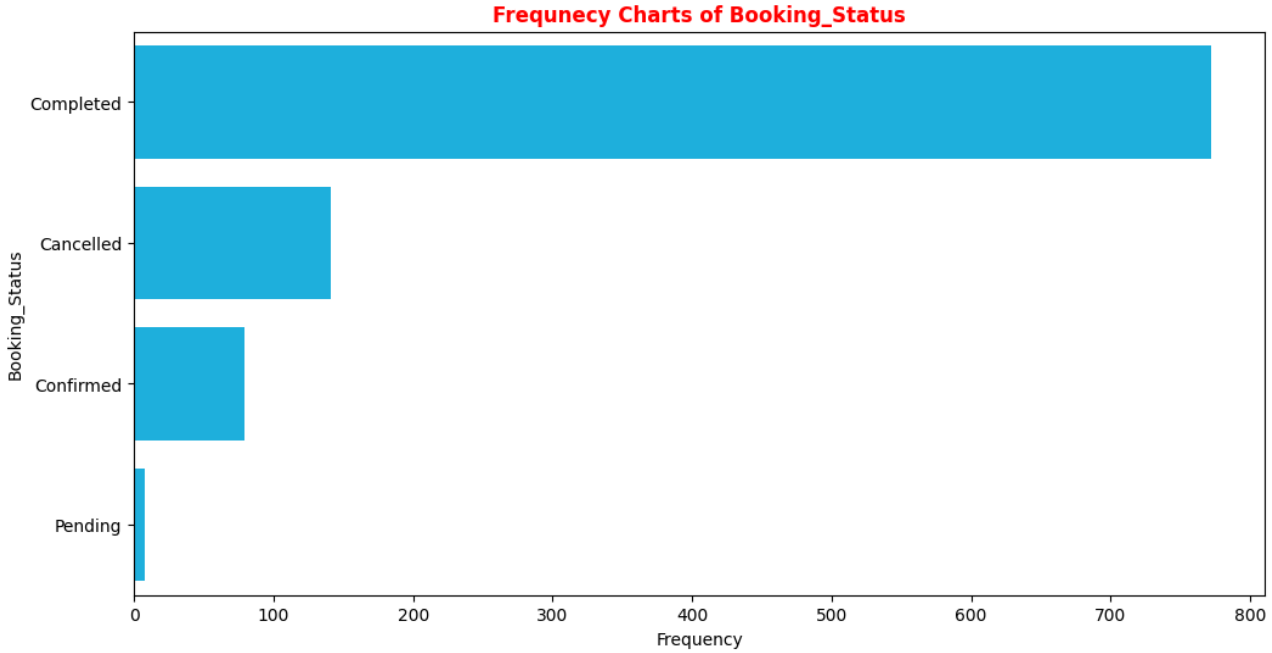
```
 Frequency of Meal_Plan:
 Meal_Plan
Breakfast Included    392
Room Only             220
Full Board            208
Half Board            180
Name: count, dtype: int64
```



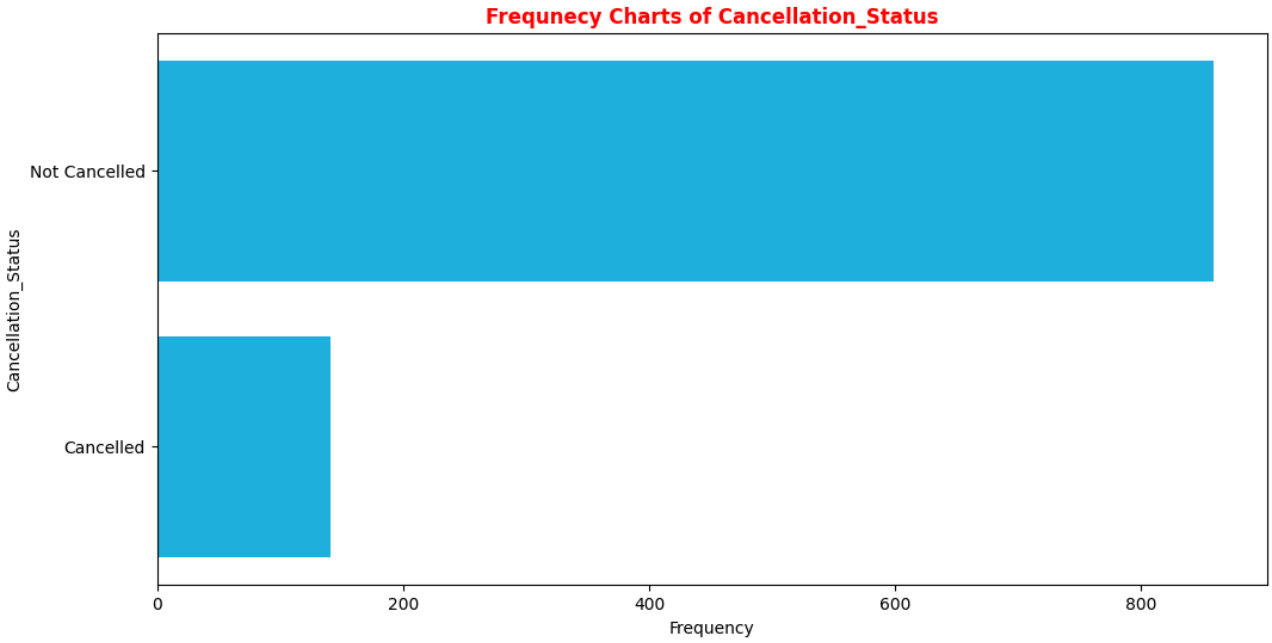

```
 Frequency of Payment_Method:
 Payment_Method
UPI               297
EMI               152
Credit Card       150
Digital Wallet    149
Debit Card        131
Net Banking       121
Name: count, dtype: int64
```




```
 Frequency of Booking_Status:
 Booking_Status
Completed    772
Cancelled    141
Confirmed     79
Pending        8
Name: count, dtype: int64
```



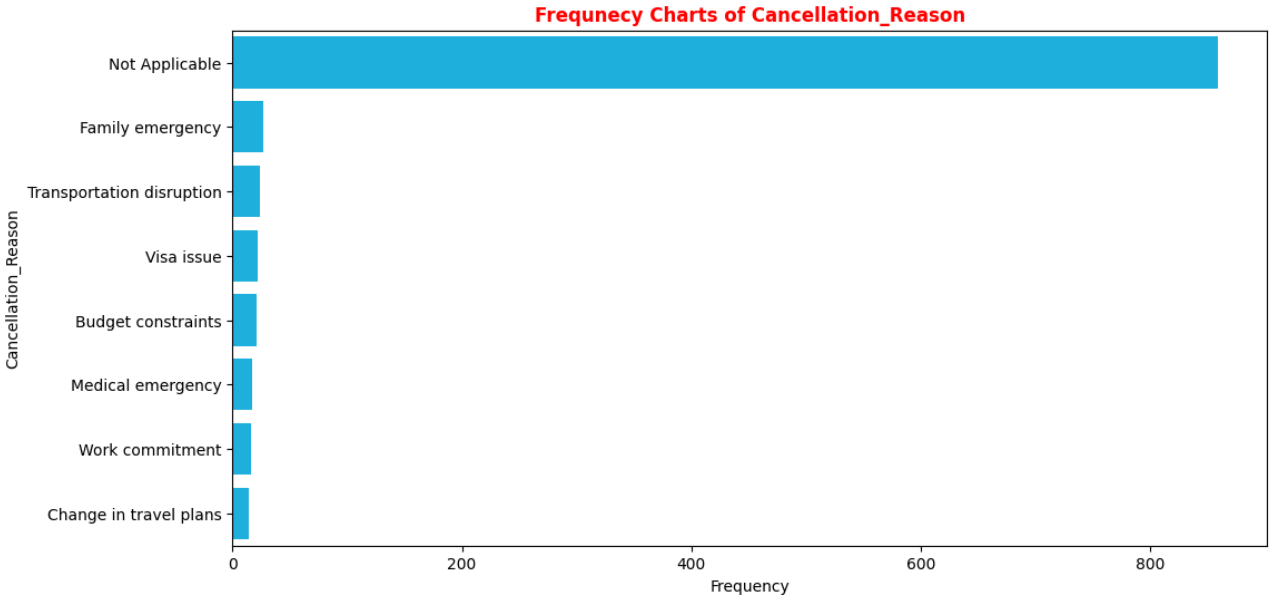

```
 Frequency of Cancellation_Status:
 Cancellation_Status
Not Cancelled    859
Cancelled        141
Name: count, dtype: int64
```



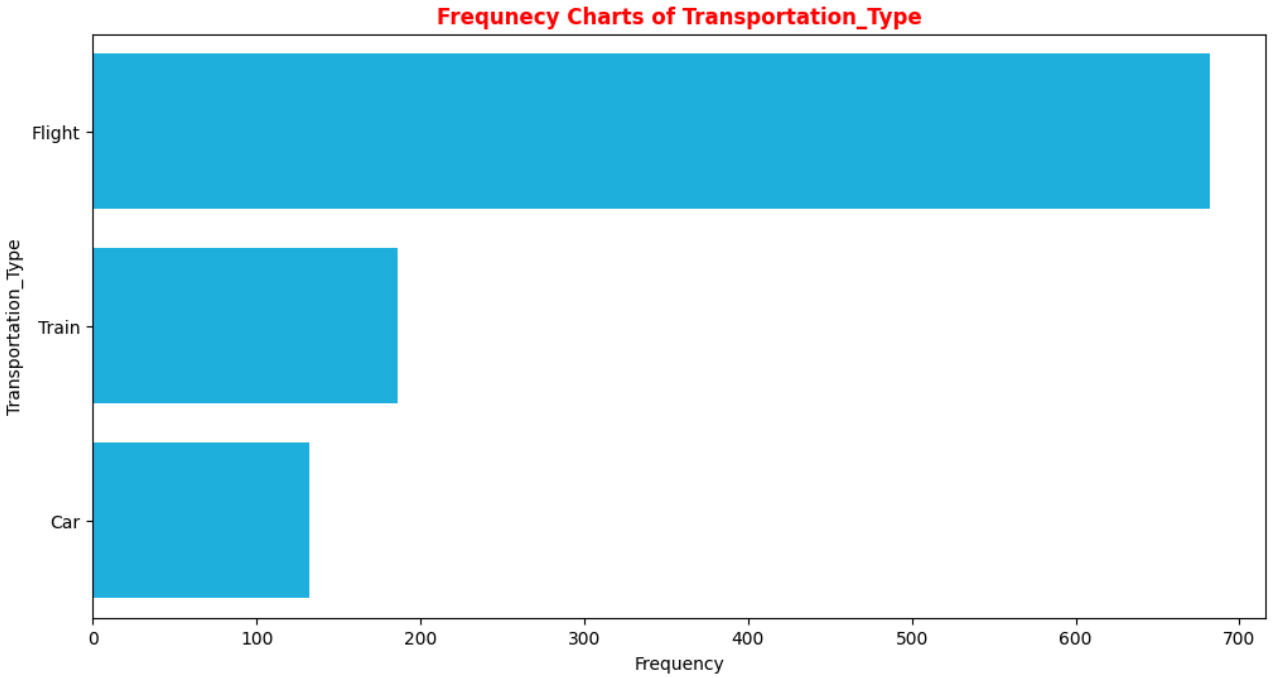
Frequency of Cancellation_Reason:

Cancellation_Reason	
Not Applicable	859
Family emergency	27
Transportation disruption	24
Visa issue	22
Budget constraints	21
Medical emergency	17
Work commitment	16
Change in travel plans	14

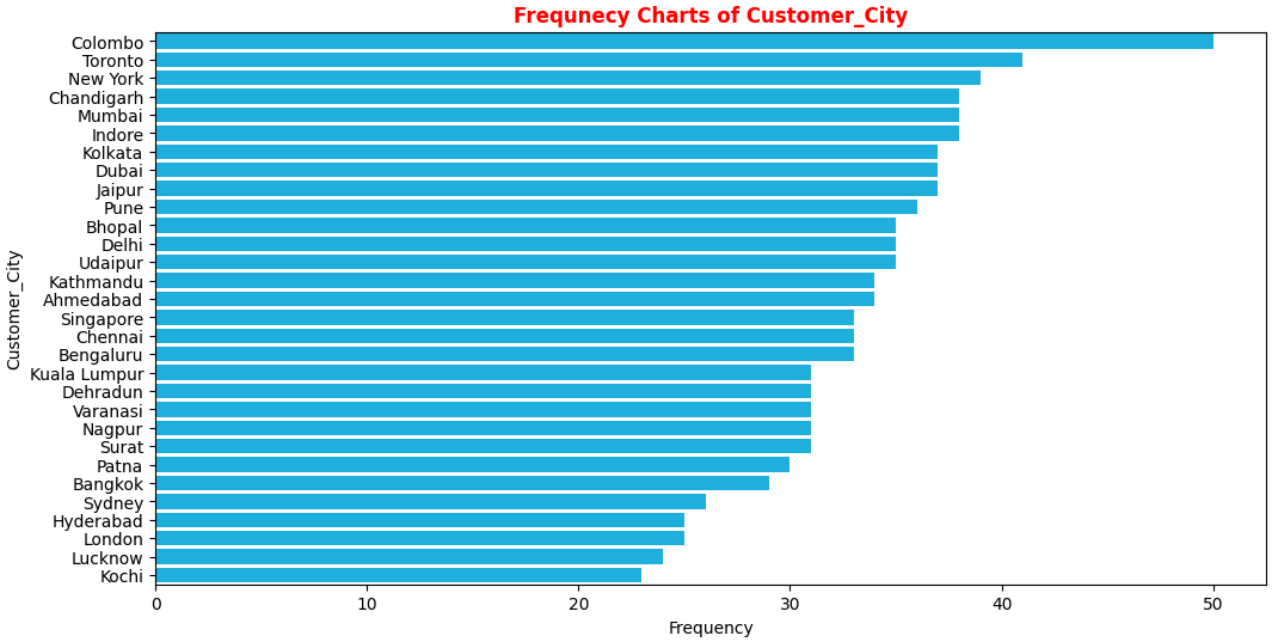
Name: count, dtype: int64



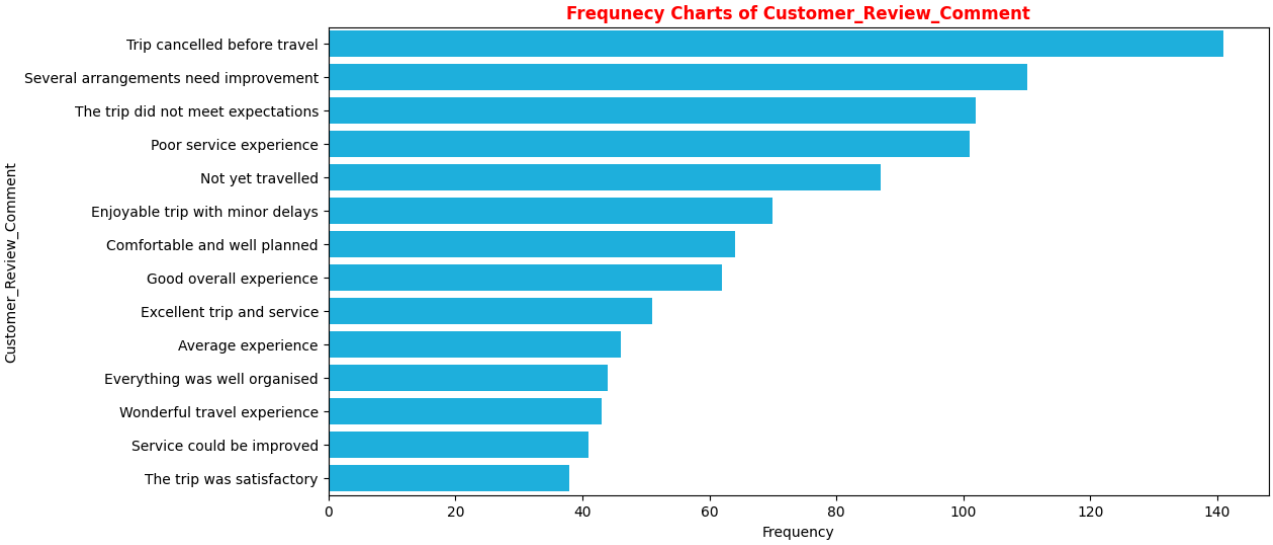
```
 Frequency of Transportation_Type:
 Transportation_Type
Flight    682
Train     186
Car       132
Name: count, dtype: int64
```



```
 Frequency of Customer_City:
 Customer_City
Colombo          50
Toronto          41
New York         39
Chandigarh       38
Mumbai           38
Indore           38
Kolkata          37
Dubai            37
Jaipur           37
Pune             36
Bhopal           35
Delhi            35
Udaipur          35
Kathmandu        34
Ahmedabad        34
Singapore        33
Chennai          33
Bengaluru        33
Kuala Lumpur     31
Dehradun         31
Varanasi         31
Nagpur           31
Surat            31
Patna            30
Bangkok          29
Sydney           26
Hyderabad        25
London           25
Lucknow          24
Kochi            23
Name: count, dtype: int64
```



```
 Frequency of Customer_Review_Comment:
 Customer_Review_Comment
Trip cancelled before travel              141
Several arrangements need improvement     110
The trip did not meet expectations        102
Poor service experience                   101
Not yet travelled                          87
Enjoyable trip with minor delays           70
Comfortable and well planned               64
Good overall experience                    62
Excellent trip and service                 51
Average experience                         46
Everything was well organised              44
Wonderful travel experience                43
Service could be improved                  41
The trip was satisfactory                  38
Name: count, dtype: int64
```



Meaningful insights

The dataset contains **1,000 bookings**. The charts show these main patterns:

- **Gender:** Female customers account for **568 bookings (56.8%)**, compared with **432 (43.2%)** for male customers. The customer mix is somewhat female-skewed.
- **Customer country: India dominates** with **655 bookings (65.5%)**. Sri Lanka is a distant second with 50; each other country contributes 41 or fewer. The bookings are therefore heavily concentrated in India.
- **Meal plan: Breakfast Included** is the most common option (**392; 39.2%**). Room Only, Full Board, and Half Board are less frequent and relatively close to one another.
- **Payment method: UPI** is the most-used method (**297; 29.7%**). EMI,

credit card, and digital wallet are each around 15%; debit card and net banking are less common.

- **Booking status:** Most bookings are **Completed (772; 77.2%)**. There are 141 cancelled bookings, 79 confirmed bookings, and only 8 pending bookings.
- **Cancellation status: 859 bookings (85.9%) were not cancelled**, while **141 (14.1%) were cancelled**. This aligns with the cancelled count in Booking Status.
- **Cancellation reasons:** Among the 141 cancellations, **family emergency** is the most frequent reason (27; about 19% of cancellations), followed by transportation disruption (24), visa issues (22), and budget constraints (21). No single reason accounts for most cancellations.
- **Transportation: Flights** are the clear preference (**682; 68.2%**), followed by train (186; 18.6%) and car (132; 13.2%).
- **Customer city: Colombo** has the most bookings (50), followed by Toronto (41) and New York (39). Counts across cities are much more evenly distributed than across countries, with the leading city representing just 5% of bookings.
- **Review comments:** “Trip cancelled before travel” is the most common comment (141), matching the number of cancelled bookings. Other frequent comments mention arrangements needing improvement (110), trips not meeting expectations (102), and poor service (101). Positive comments also appear, such as “Enjoyable trip with minor delays” (70) and “Good overall experience” (62), suggesting feedback is mixed.

Overall: The bookings are strongly concentrated in India, flights are the most common transportation choice, and most bookings are completed and not cancelled. The comment counts point to service and arrangement quality as possible areas for attention. These are frequency patterns only; the charts do not establish why customers chose a particular option or whether one factor caused another.

In [15…

```
# Date Distribution
date_columns = ['Travel_Date', 'Return_Date']
for col in date_columns:
    print(f"\n Date Range of {col}: {Travel[col].min()} to {Travel[col].m
    print(f"\n Time Spent: {(Travel[col].max() - Travel[col].min()).days}
# Month Distribution
plt.figure(figsize=(13,6))
sns.countplot(x = Travel['Travel_Date'].dt.month_name(),color = '#007FFF'
plt.title('Distribution of Months', fontsize = 12, fontweight = 'bold', c
plt.ylabel('Frequency')
plt.xticks(rotation = 45)
```

```
plt.show()

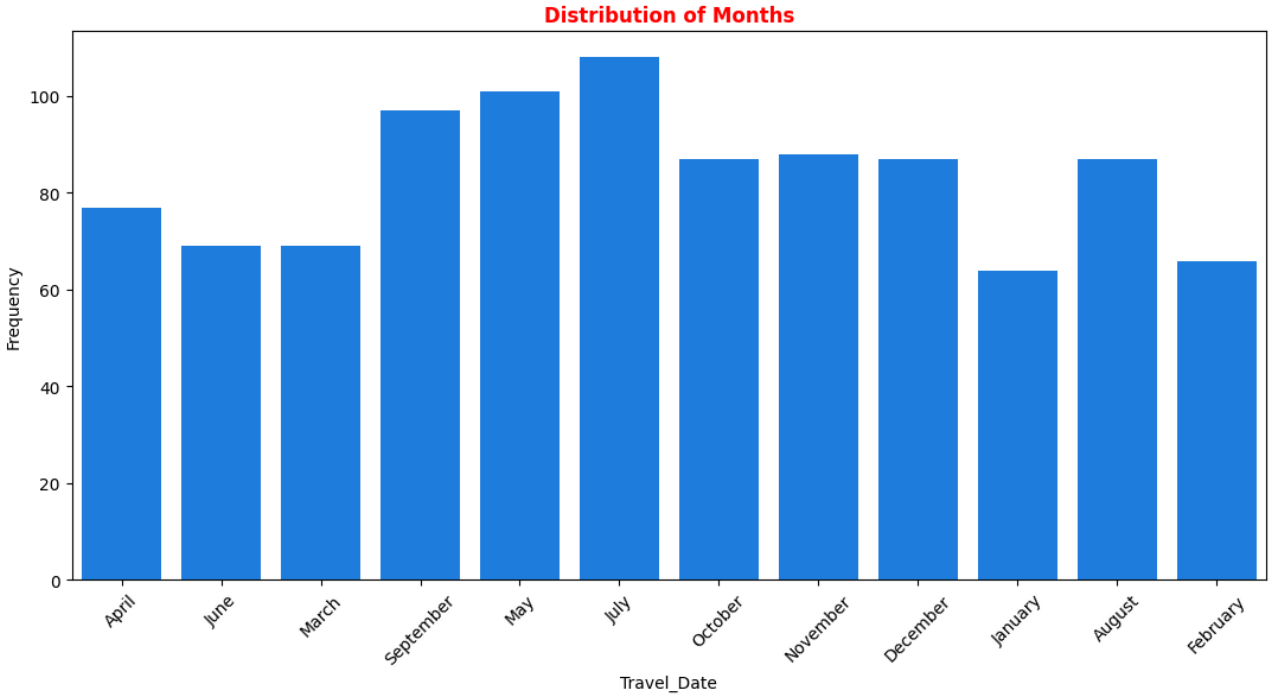
# Years Distribution
plt.figure(figsize=(6,4))
sns.countplot(x = Travel['Travel_Date'].dt.year,color = '#007FFF')
plt.title('Distribution of Years', fontsize = 12, fontweight = 'bold', co
plt.ylabel('Frequency')
plt.xticks(rotation = 45)
plt.show()
```

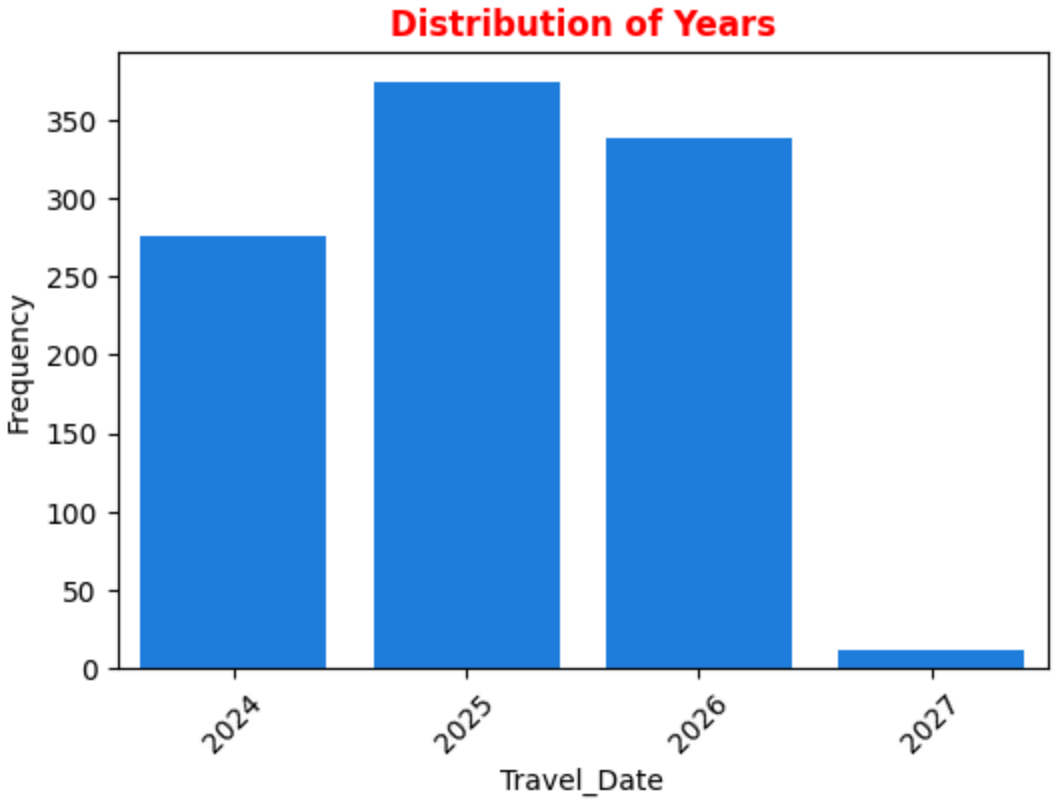
```
Date Range of Travel_Date: 2024-01-31 00:00:00 to 2027-02-14 00:00:00

Time Spent: 1110 days

Date Range of Return_Date: 2024-02-04 00:00:00 to 2027-02-22 00:00:00

Time Spent: 1114 days
```





Meaningful Insights

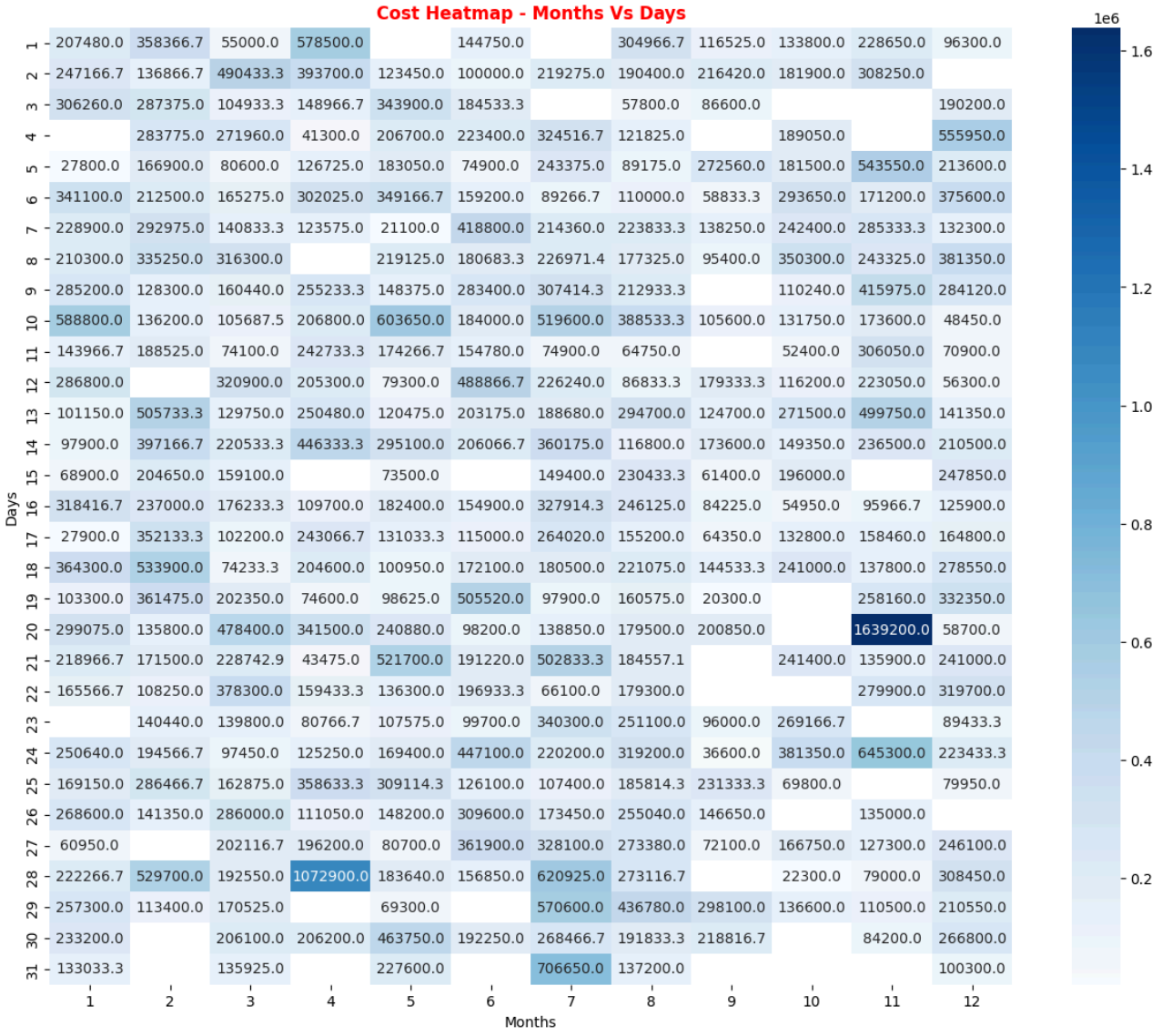
- The dataset covers a long period of travel activity, from 2024-01-31 to 2027-02-14 for Travel_Date and from 2024-02-04 to 2027-02-22 for Return_Date. This means the tourism records span nearly 3 years, giving a strong time-based view of travel patterns.
- The overall time span is very similar for both travel and return dates, which suggests the trip records are consistent and well-structured. There is no major mismatch between the start and end of travel cycles.
- The year-wise distribution shows that travel activity is not uniform across years. The busiest years appear to be 2025 and 2026, while 2024 and 2027 are comparatively lower. This could indicate either a true rise in tourism demand over time or partial-year coverage in the early and late years.
- The monthly distribution reveals a clear seasonal pattern. Travel demand peaks in certain months and dips in others, which strongly suggests tourism is influenced by seasonality. The busiest periods appear around the middle and later parts of the year, which often align with holiday and vacation travel.

- The bar chart indicates that tourists are not spread evenly across the year; instead, there are recurring high-demand months and lower-demand months. This is valuable for planning promotions, pricing, hotel occupancy, and transport scheduling.
- The visualization suggests that tourism demand is more active during favorable weather periods rather than during colder months, meaning climate and seasonal conditions likely drive travel behavior.
- Overall, the data reflects a stable tourism pattern with pronounced seasonality and a strong annual trend, making it useful for forecasting demand, managing resources, and designing targeted marketing campaigns.

In short: tourism activity is seasonal, concentrated in specific months, and strongest in the middle years of the dataset, which highlights the importance of time-based planning and demand forecasting.

In [16…

```
heatmapData = Travel.pivot_table(index = Travel['Booking_Date'].dt.day,
columns= Travel['Booking_Date'].dt.month,
values = 'Total_Trip_Cost', aggfunc = 'mean' )
plt.figure(figsize=(15,12))
sns.heatmap(heatmapData, annot = True,
 fmt = '.1f', cmap="Blues")
plt.title("Cost Heatmap - Months Vs Days",
fontsize = 12, fontweight = 'bold', color = 'red')
plt.ylabel("Days")
plt.xlabel('Months')
plt.show()
```



Meaningful Insights

- The cost is not uniform across the year; it varies significantly by both month and day. This means travel or operational costs are highly time-sensitive rather than constant.
- There are clear peak-cost clusters in specific month-day combinations. These spikes suggest seasonal surges, special events, demand peaks, or unusually high activity during certain dates.
- The heatmap shows that some months are consistently more expensive than others, while other periods remain relatively low. This indicates a seasonal pattern in costs.
- A few isolated high-cost cells stand out against surrounding values, which

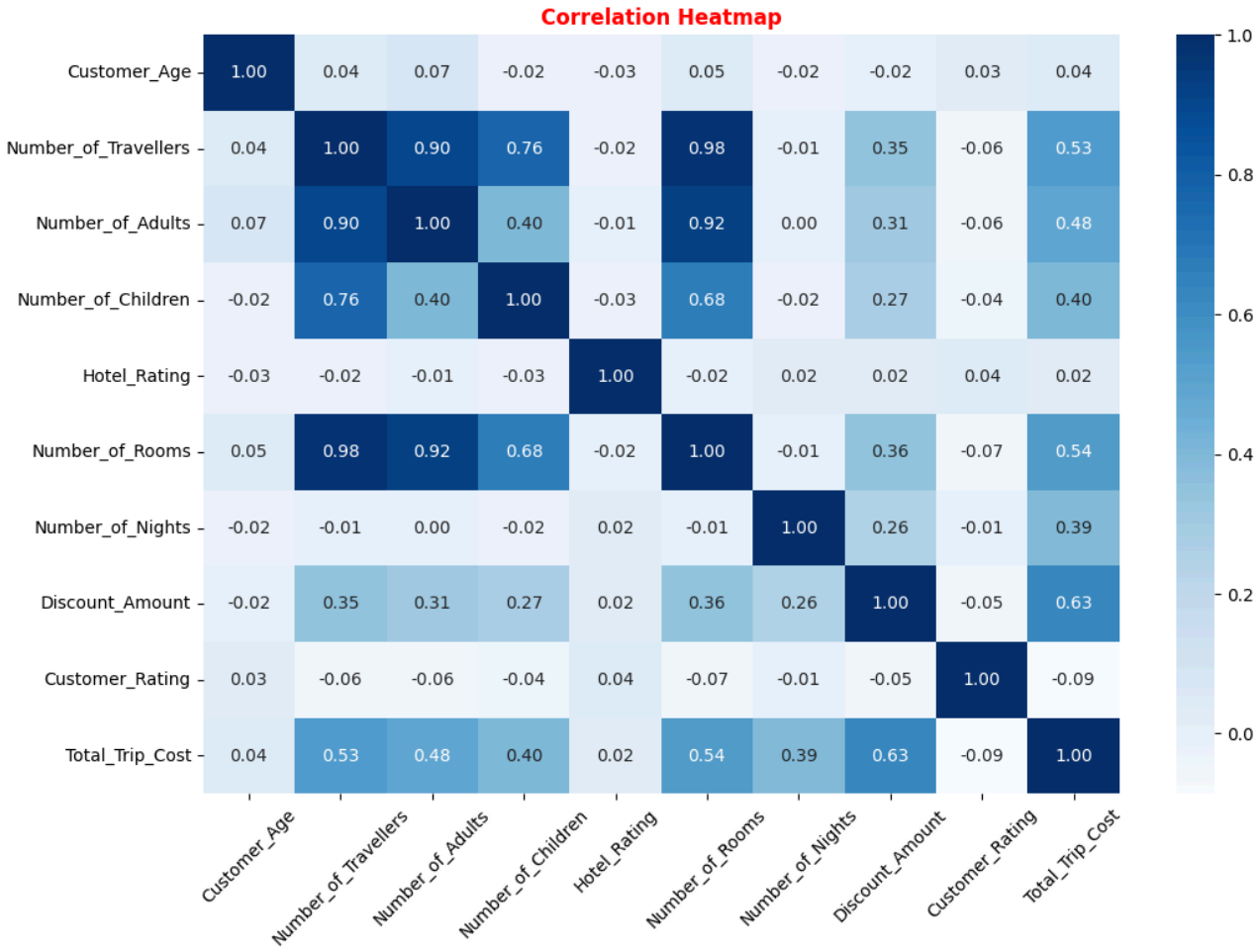
could represent:
    - holiday travel periods,
    - peak booking demand,
    - weather-driven tourism surges,
    - supply shortages or premium pricing.
- The variation within the same month is also important: even within one month, some days are much more expensive than others. This means planning should be date-specific, not just month-based.
- The darker blue areas represent high-cost periods, while lighter cells indicate lower-cost periods. The contrast suggests that cost management can be improved by identifying and preparing for these high-cost windows in advance.
- The pattern likely reflects seasonality in tourism demand, where certain months attract higher visitor volume and therefore higher associated costs.
- For decision-making, the most valuable insight is that cost planning should focus on these high-intensity periods, since they can substantially increase expenses if not anticipated.

Overall takeaway:

- Tourism-related costs follow a seasonal and date-specific pattern.
- High-cost dates are concentrated in certain periods, which means better forecasting, pricing, and budgeting can be done by targeting those peak windows.

In [17…

```
#Corelation Analysis
Corelation = Travel[Num].corr()
plt.figure(figsize = (12,8))
sns.heatmap(Corelation, annot = True,
 fmt = '.2f', cmap = 'Blues')
plt.title('Correlation Heatmap',fontsize =12,
color = 'red', fontweight ='bold')
plt.xticks(rotation = 45)
plt.show()
```



Meaningful Insights

- The strongest relationships are among the trip-size and accommodation variables:
  - Number_of_Travellers is strongly correlated with Number_of_Adults (0.90)
  - Number_of_Travellers is also strongly correlated with Number_of_Rooms (0.98)
  - Number_of_Adults and Number_of_Rooms are highly correlated (0.92)
  - This indicates that larger groups usually book more rooms and travel together, which is expected in tourism and hotel bookings.
- Group size is a major factor in trip cost:
  - Total_Trip_Cost has moderate positive correlation with Number_of_Rooms (0.54), Number_of_Travellers (0.53), and Number_of_Adults (0.48)

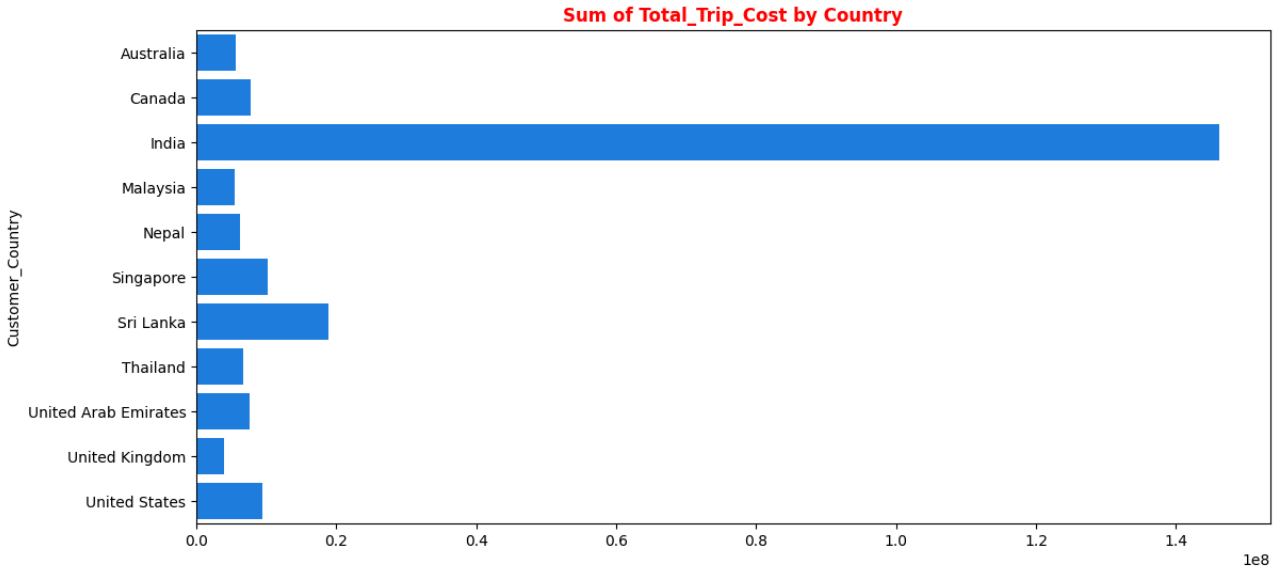
    - This shows that higher trip cost is mainly driven by more travelers and room requirements, rather than by age or other demographics.
- Children contribute to trip size but less strongly than adults:
    - Number_of_Children has positive but weaker links with Number_of_Travellers (0.76) and Number_of_Rooms (0.68)
    - This suggests adult group size is a stronger driver of accommodation planning than the presence of children.
- Discount amount is moderately related to trip cost:
    - Discount_Amount and Total_Trip_Cost are positively correlated (0.63)
    - Larger trips may attract larger discounts, or discount decisions may be linked to booking size and package value.
- Customer age is not a strong predictor of travel pattern or trip cost:
    - Customer_Age shows almost no meaningful correlation with the other variables
    - This suggests age does not strongly determine trip size, room count, or total spending in this dataset.
- Hotel rating is not strongly linked to trip characteristics:
    - Hotel_Rating shows near-zero correlation with travel size, room count, trip cost, and customer ratings
    - This may imply that hotel quality is not strongly tied to booking size or customer age in this dataset.
- Customer satisfaction is weakly related to trip size and cost:
    - Customer_Rating has weak or near-zero correlations with most variables
    - This means customer satisfaction is not strongly explained by group size, room count, or total trip spending alone.

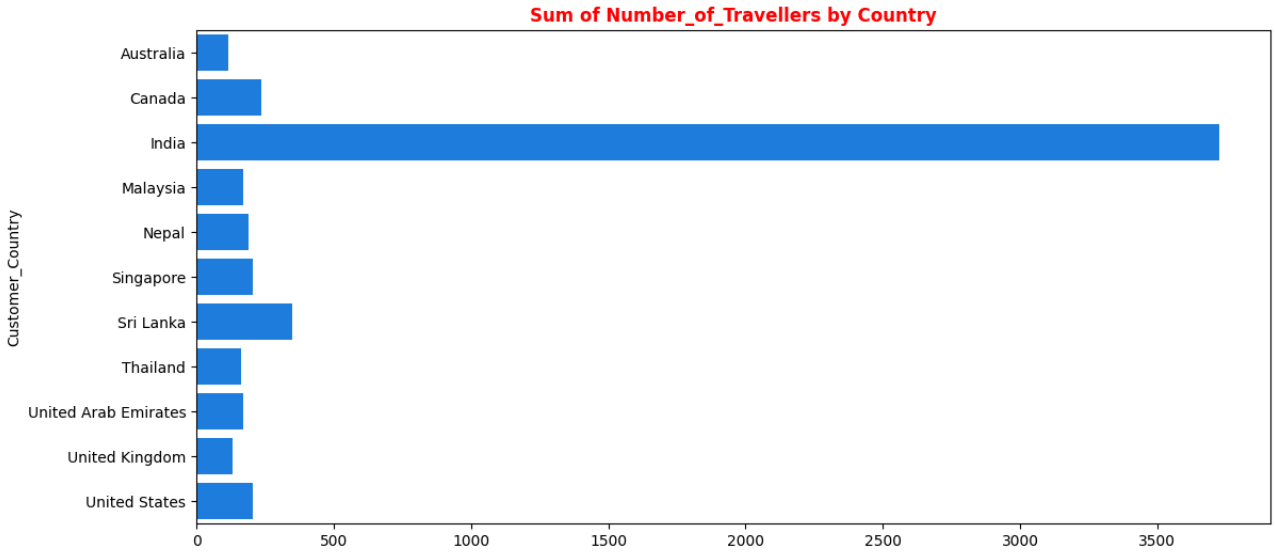
Overall conclusion:

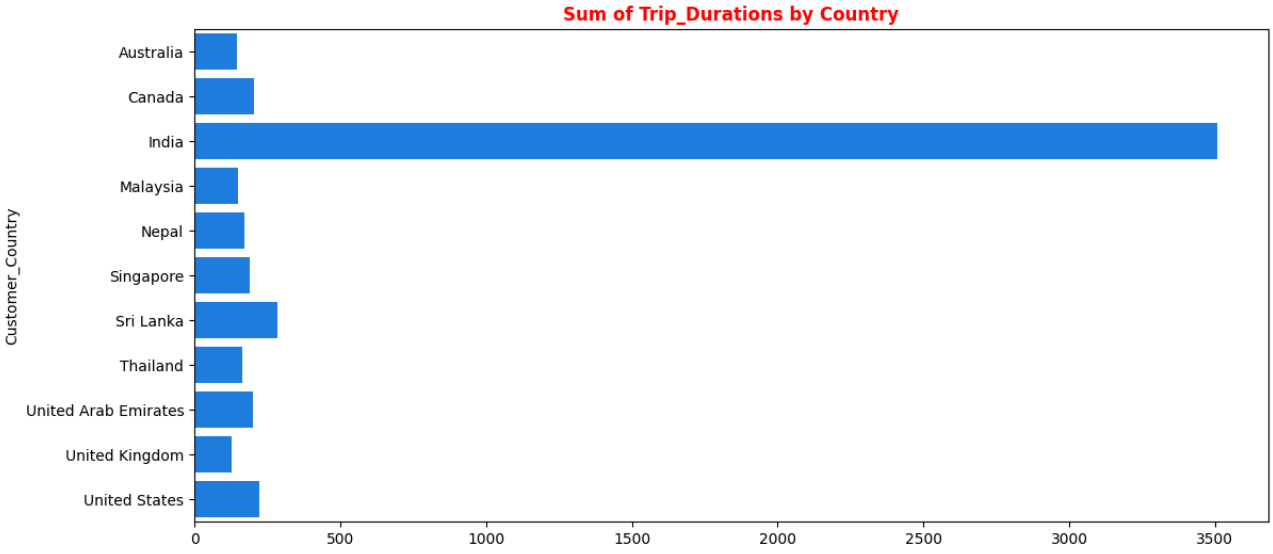
- The most important drivers of tourism spending are group size and accommodation needs, especially Number_of_Travellers, Number_of_Adults, and Number_of_Rooms.
- Customer age and hotel rating are not major factors in explaining trip cost or travel behavior in this dataset.
- The data suggests that pricing and trip planning are more influenced by group size and logistics than by demographic characteristics.

In [18…

```
# Country Wise Analysis
Travel['Trip_Durations'] = (Travel['Return_Date'] - Travel['Travel_Date']
Country = (Travel.groupby('Customer_Country')
           [['Total_Trip_Cost','Number_of_Travellers'
             ,'Trip_Durations']].sum().reset_index())
Country_Col = ['Total_Trip_Cost','Number_of_Travellers'
             ,'Trip_Durations']
for col in Country_Col:
    plt.figure(figsize = (13,6))
    sns.barplot(y = Country['Customer_Country'],
    x = Country[col],color = "#007FFF")
    plt.title(f'Sum of {col} by Country',
    fontsize = 12, color = 'red', fontweight = 'bold')
    plt.xlabel(' ')
    plt.show()
```







meaningful insights

- India stands out as the dominant market in all three charts:
  - It has the highest total trip cost
  - It has the highest total number of travelers
  - It has the highest total trip duration
- This means India is the largest contributor to tourism demand in the dataset, and it likely drives the biggest share of travel spending and trip activity.
- The three charts show a consistent pattern:
  - Countries with more travelers also tend to have higher total trip costs and longer trip durations.
  - This suggests that travel volume and trip length are major drivers of overall spending.
- United States, United Kingdom, and United Arab Emirates appear to be significant secondary markets, but still below India.
- Countries like Australia, Canada, Malaysia, Nepal, Singapore, Sri Lanka, and Thailand show comparatively lower totals, indicating smaller tourism contribution or shorter/less frequent trips.
- The data suggests a highly uneven market distribution:
  - A few countries contribute most of the travel activity.
  - The tourism demand is concentrated rather than evenly spread across

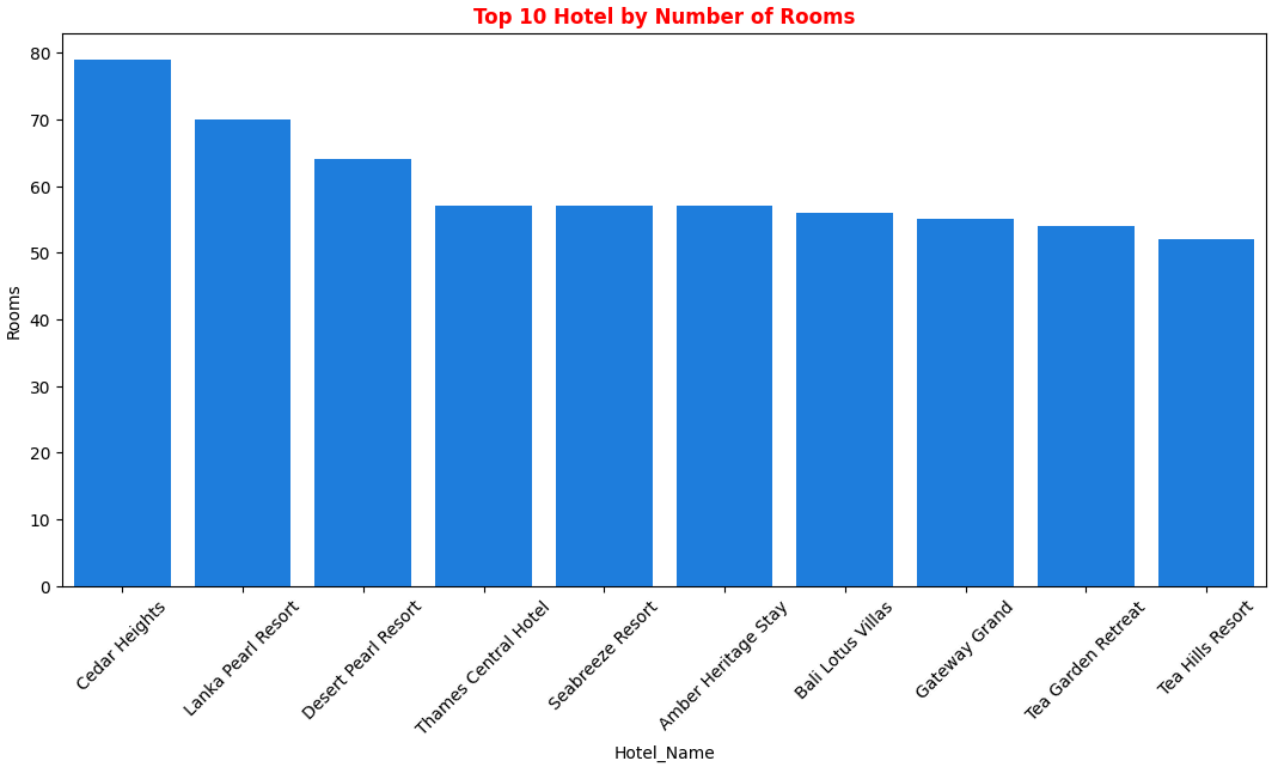
all countries.
    - This is useful for resource allocation, pricing strategy, and targeted marketing.
- The total trip duration pattern aligns with traveler volume:
    - Countries with more travelers generally also have longer cumulative travel time.
    - This implies that longer or more frequent trips are strongly linked to larger market engagement.
- Overall insight:
    - India is the strongest source market.
    - The market is concentrated in a small number of countries.
    - Travel spending and trip duration increase with travel volume, making group size and trip frequency key indicators for tourism performance.

This kind of analysis is valuable for:

- focusing marketing efforts on the highest-performing countries,
- forecasting future demand,
- allocating travel packages and operational resources,
- designing country-specific pricing and promotions.

In [19…

```
#per counrty by avg cost, trip duration, avg travellers per hotel by avg
Hotel = (Travel.groupby('Hotel_Name')['Number_of_Rooms'].
         sum().reset_index(name = 'Rooms').
         sort_values(by= 'Rooms', ascending = False)[0:10])
plt.figure(figsize = (13,6))
sns.barplot(x = Hotel['Hotel_Name'],
y= Hotel['Rooms'],color = '#007FFF')
plt.title('Top 10 Hotel by Number of Rooms',
 fontsize = 12, fontweight = 'bold', color = 'red')
plt.xticks(rotation = 45)
plt.show()
```



Meaningful insight

- Cedar Heights is the clear leader with the highest room capacity, at about 80 rooms. This suggests it has the strongest accommodation scale in the set.
- The next two hotels, Lanka Pearl Resort and Desert Pearl Resort, also have very high capacities, around 70–75 rooms. This creates a top-tier cluster of large properties with significant inventory.
- The chart shows a clear hierarchy: the top 3 hotels dominate the market in terms of room count, while the remaining properties are smaller but still substantial, ranging roughly from 40 to 60 rooms.
- The distribution is fairly balanced among mid-size hotels, which means the market is not dominated by a single hotel alone but by a few large players plus several medium-sized competitors.
- This indicates that the tourism market is moderately concentrated, where a small group of hotels holds a large share of room inventory. These properties likely have stronger bargaining power, better occupancy potential, and more resources for marketing and operations.
- Smaller hotels in the lower half of the chart still have meaningful capacity, suggesting they can compete by focusing on niche services, unique experiences, premium hospitality, or strategic locations rather than scale

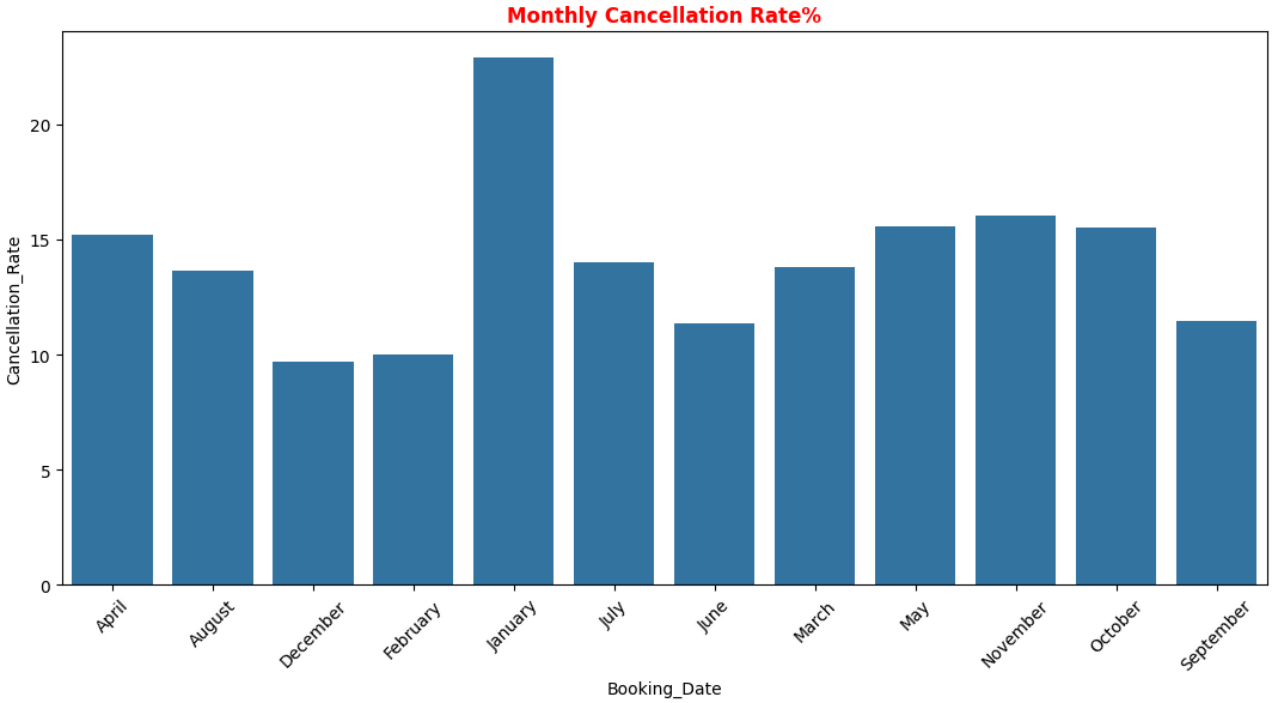
alone.

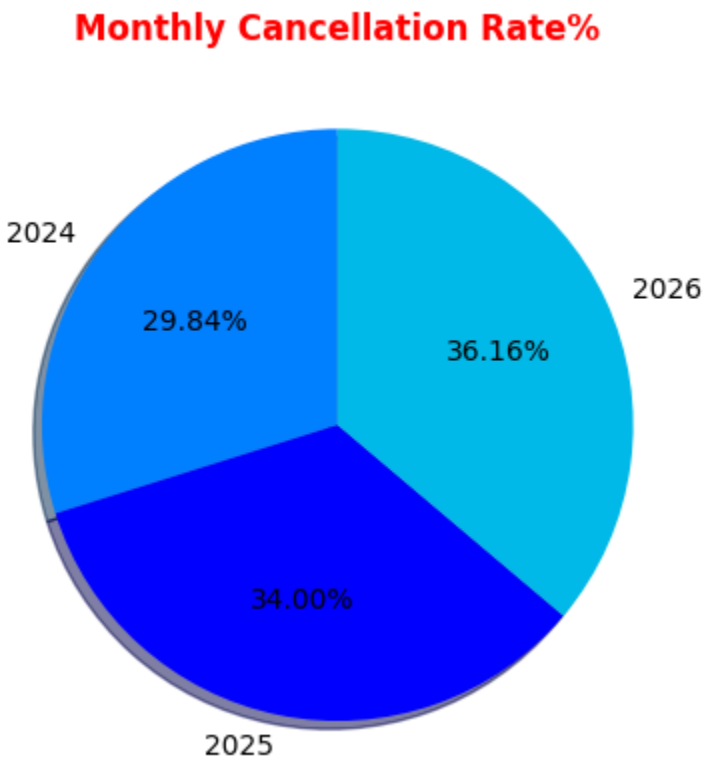
- Overall, the chart suggests that the area has a strong mix of large and medium-sized accommodation options, with a few dominant hotels leading the room inventory while many others serve as competitive alternatives.

In [20…

```
#Cancelation By Booking Date
Sort_Travel= Travel.sort_values(by = 'Booking_Date')
Sort_Travel['Is_Cancelled'] = (Sort_Travel['Cancellation_Status']=='Cance
#Monthly Cancelation Rate
Monthly_Cancellation = (Sort_Travel.groupby(Sort_Travel['Booking_Date'].
dt.month_name()).agg( Cancellation_Rate =
                     ('Is_Cancelled', 'mean'))).reset_index()
Monthly_Cancellation['Cancellation_Rate'] = (
    Monthly_Cancellation['Cancellation_Rate']*100
    ).round(2)
plt.figure(figsize = (13,6))
sns.barplot(x = Monthly_Cancellation['Booking_Date'],
y= Monthly_Cancellation['Cancellation_Rate'])
plt.title("Monthly Cancellation Rate%", color = "red",
    fontsize = 12, fontweight = 'bold')
plt.xticks(rotation = 45)
plt.show()

# Yearly cancelation rate
Yearly_Cancellation = (Sort_Travel.groupby(Sort_Travel['Booking_Date'].
dt.year).agg(Cancellation_Rate =
('Is_Cancelled', 'mean'))).reset_index()
Yearly_Cancellation['Cancellation_Rate'] = (
    Yearly_Cancellation['Cancellation_Rate']*100
    ).round(2)
plt.pie(Yearly_Cancellation['Cancellation_Rate'],
        labels=Yearly_Cancellation['Booking_Date'],
        autopct = '%1.2f%%', startangle = 90,
        shadow = True, colors = ['#007FFF','#0000FF','#00B9E8'])
plt.title("Monthly Cancellation Rate%", color = "red",
    fontsize = 12, fontweight = 'bold')
plt.show()
```





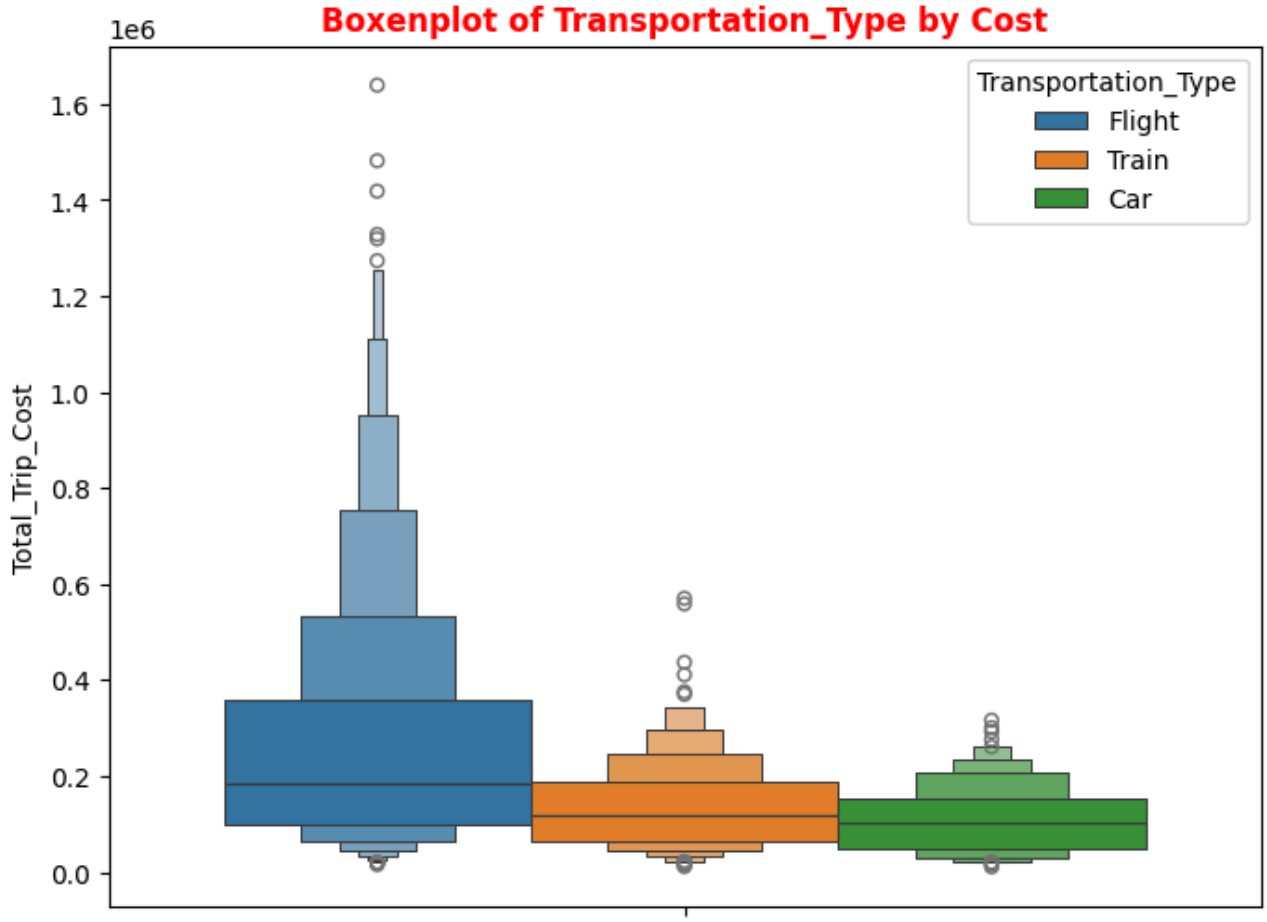
Meaningful insights

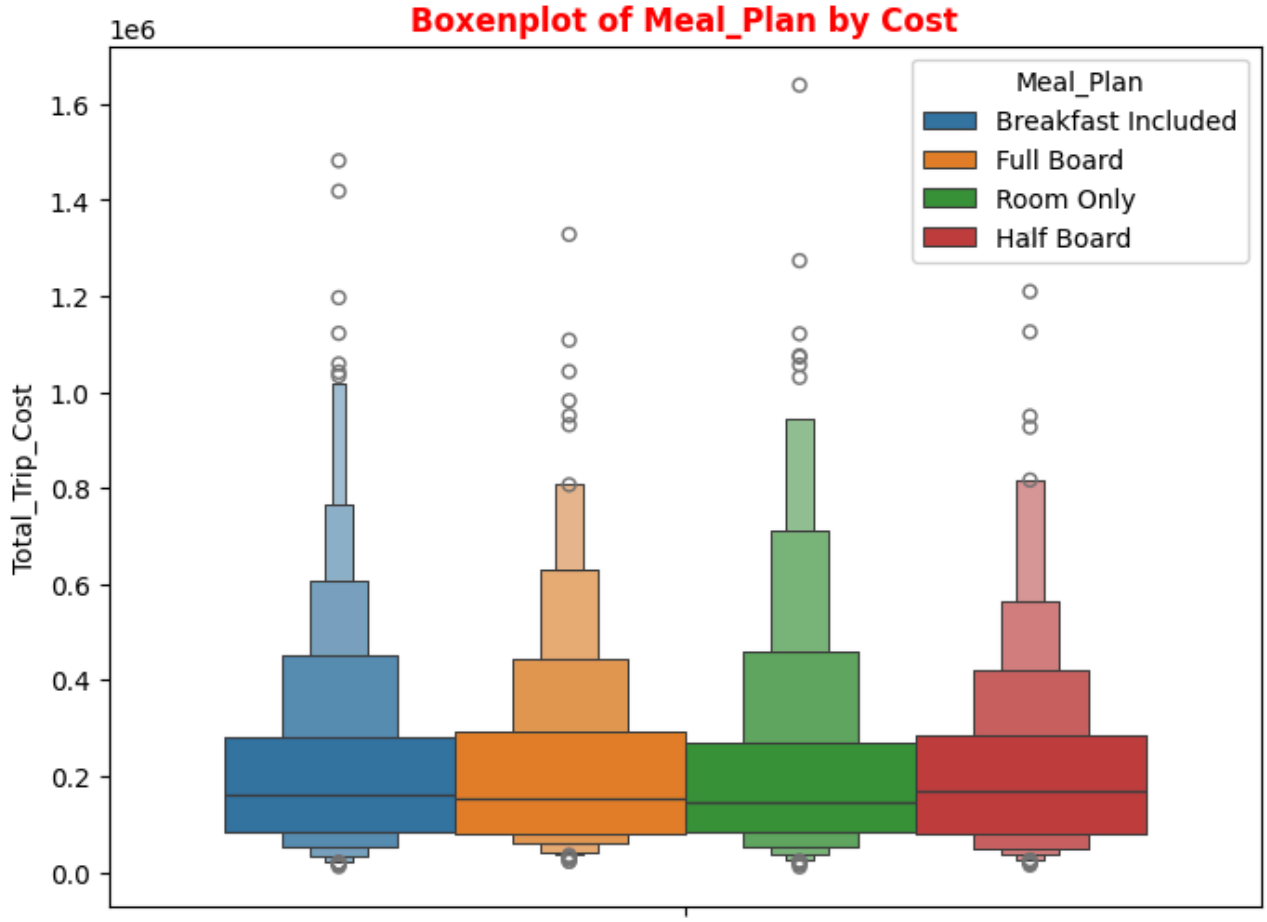
- **January has the highest monthly cancellation rate**, at about **23%**—roughly twice December's rate.

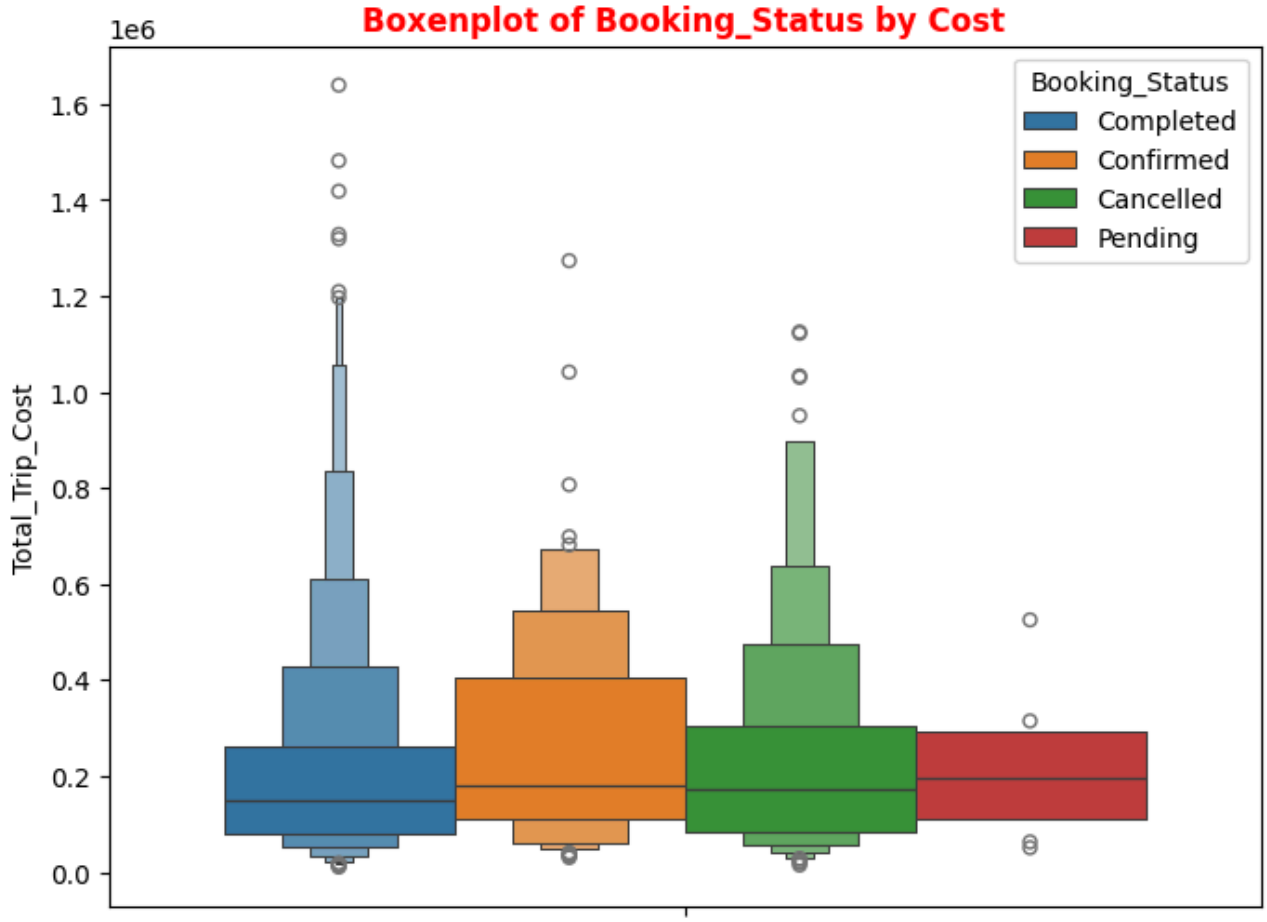
- **December is the lowest**, at about **10%**. February and September are also relatively low, around **10–12%**.
- Most other months fall between **14% and 16%**, suggesting cancellations are generally fairly steady outside January's peak and the lower-rate months.
- The yearly rates appear to rise from **29.84% in 2024** to **34.00% in 2025** and **36.16% in 2026**. This may indicate a worsening trend, but compare cautiously if 2026 contains only partial-year data.

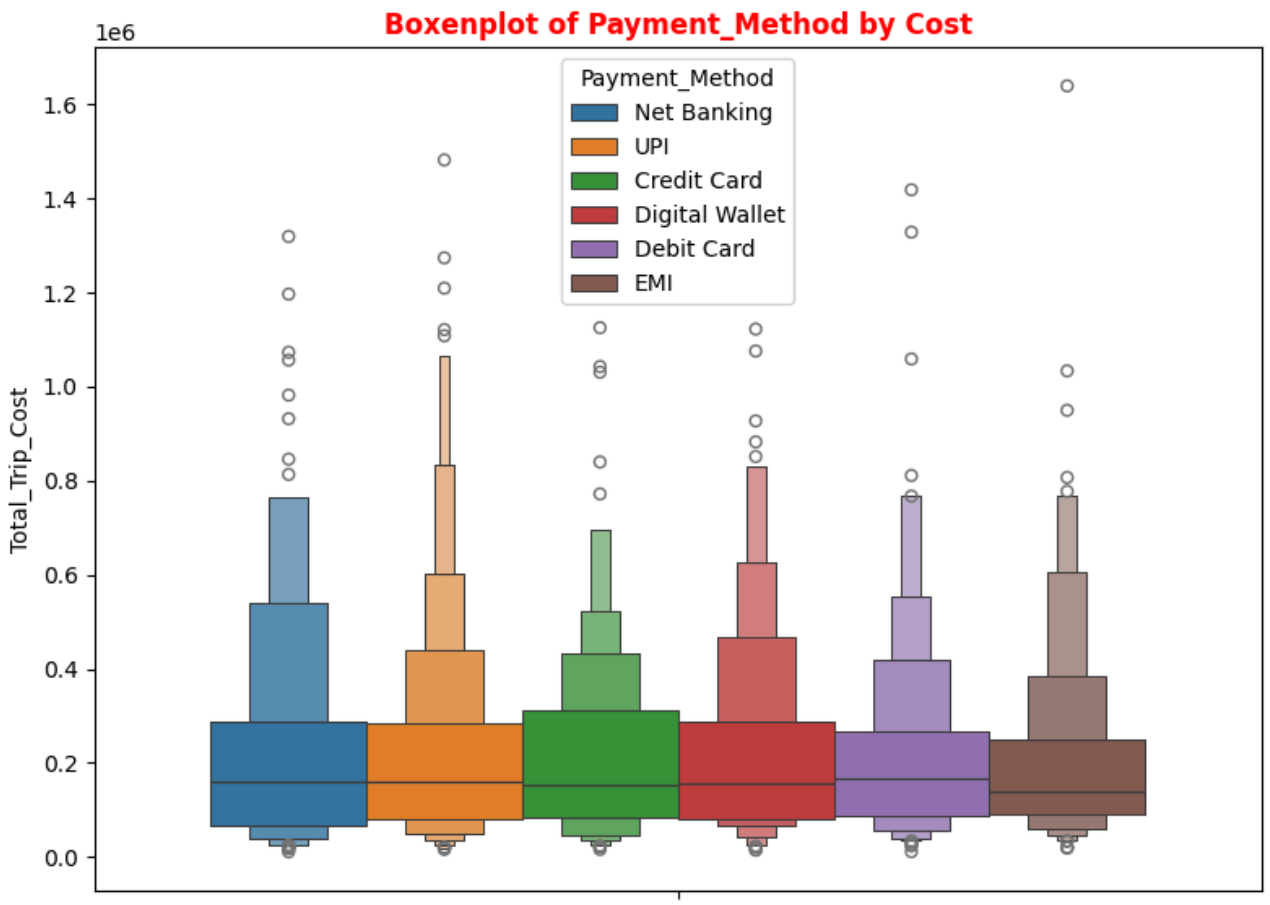
In [21…

```
#other Cost in Travel
Other = ['Transportation_Type','Meal_Plan',
'Booking_Status', 'Payment_Method']
for x, col in enumerate(Other):
    plt.figure(figsize = (8,6))
    sns.boxenplot(y = Travel['Total_Trip_Cost'],
     hue = Travel[col])
    plt.xlabel(' ')
    plt.title(f'Boxenplot of {col} by Cost', fontsize = 12,
    fontweight = 'bold',color = 'red' )
plt.tight_layout()
plt.show()
```


Boxenplot of Transportation_Type by Cost
1e6
Total_Trip_Cost
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Transportation_Type
Flight
Train
Car


Boxenplot of Meal_Plan by Cost
1e6
Total_Trip_Cost
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Meal_Plan
Breakfast Included
Full Board
Room Only
Half Board


Boxenplot of Booking_Status by Cost
1e6
Total_Trip_Cost
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Booking_Status
Completed
Confirmed
Cancelled
Pending



Insights: Travel Cost by Booking Characteristics

- **Transportation type:** Flights have the highest median trip cost and the widest spread, with several very high-cost outliers. Train and car trips are generally less expensive, and their cost distributions are more compact.
- **Meal plan:** Median costs are broadly similar across meal plans. The distributions overlap considerably, although some plans show a wider range of costs and high-cost outliers. The chart does not suggest a clear, consistent cost increase from including meals.
- **Booking status:** Costs overlap across completed, confirmed, cancelled, and pending bookings. Their medians are relatively close, so booking status alone does not appear to strongly distinguish trip cost.
- **Payment method:** Costs also overlap across payment methods, with high-cost outliers in several groups. There is no clear evidence from this plot that a particular payment method is associated with higher or lower trip costs.
- **Overall:** Transportation type shows the clearest relationship with trip

cost; flights tend to be more expensive. The other categories have substantial overlap, so their apparent differences should be interpreted cautiously. The high-cost outliers across the plots indicate that a small number of trips are much more expensive than typical bookings.

Hypothesis Analysis

TTest

Is there a significant difference in average Travel Cost between Male and Female customers?

- **H0: There is no significant difference in average Travel Cost On Male and Female**
- **H1: There is a significant difference in average Travel Cost on Male and Female**

In [29…

```
MaleCustomer = Travel[Travel['Gender']=='Male']['Total_Trip_Cost']
FemaleCustomer = Travel[Travel['Gender']=='Female']['Total_Trip_Cost']
t_stat, p_value = st.ttest_ind(MaleCustomer, FemaleCustomer, equal_var =
print('T-Test: Travel Cost Difference Between Male and Female Customers:\
print(f"T-statistic: {t_stat:.2f}, P-value: {p_value:.2f}")
alpha = 0.05
if p_value < alpha:
    print("Reject H0")
    print("Average Cost is significantly different.")
else:
    print("Fail to reject H0")
    print("There is not enough evidence of a significant difference.")
```

```
T-Test: Travel Cost Difference Between Male and Female Customers:

T-statistic: -0.92, P-value: 0.36
Fail to reject H0
There is not enough evidence of a significant difference.
```

Here are the meaningful insights from the hypothesis test:

- The test was conducted to check whether the average travel cost differs between male and female customers.
- Null hypothesis (H0): There is no significant difference in average travel cost between male and female customers.
- Alternative hypothesis (H1): There is a significant difference.

Key result:

- T-statistic = -0.92
- P-value = 0.36
- Significance level = 0.05

Interpretation:

- Since p = 0.36 > 0.05, we fail to reject H0.
- This means there is not enough statistical evidence to say that male and female customers spend significantly different amounts on travel.
- The observed difference in average travel cost is likely due to random variation rather than a real gender-based difference.

Business insight:

- Gender alone does not appear to be a meaningful factor in determining travel cost for this dataset.
- Pricing, promotions, or travel offers may not need to be adjusted separately for male and female customers based on this metric alone.
- The analysis suggests that other factors like destination, trip duration, booking class, season, or customer segment may have a stronger influence on travel cost.

Overall conclusion:

- The average travel cost is statistically similar across male and female customers in this dataset.
- The result indicates no strong evidence of a gender-based difference in travel spending.